

ANASTASIOS KONTOGIORGIS

Data Scientist · Causal Inference & Predictive Modeling · Multi-agent Reinforcement Learning · PhD in Artificial Intelligence

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EU citizen · Based in Czechia · Open to remote positions

PROFILE

Data Scientist and AI Researcher with an upcoming PhD from Trinity College Dublin. I have 7+ years of experience across cryptocurrency security (Blockchain.com), marketing analytics and e-commerce (eBay) including two years as the sole data specialist for a 16-country online retailer. Adept in global audience segmentation, demand forecasting, and causal reasoning. I focus on rigorous academic evaluation, and designing robust machine learning systems that work with noisy, real-world data.

CORE SKILLS

Forecasting & causal measurement: Demand and revenue forecasting, campaign-uplift, and causal-impact evaluation.

Customer & product analytics: Churn modelling, propensity-based segmentation, conversion-funnel analysis, pricing analytics, recommender systems.

Statistics & experimentation: Hypothesis testing, linear and logistic regression, interaction effects, ANOVA, PCA, experimental design.

Machine learning: scikit-learn, LightGBM, PyTorch, sentence-transformer embeddings (MiniLM), FAISS vector search, LLM tooling (retrieval/embeddings), clustering, anomaly detection.

Data Engineering & MLOps: Python (pandas, NumPy, statsmodels), SQL (MySQL, SQL Server, Teradata), R, PySpark, FastAPI, Docker, model versioning and monitoring, Power BI, Tableau.

In progress (2026): AWS Certified Machine Learning – Associate

PROFESSIONAL EXPERIENCE

Research Intern University of Oslo, Department of Informatics **11/2024 – 05/2025**

Paid six-month research post, supervised by Prof. Eric Jul.

- Co-authored a systematic taxonomy and comprehensive field Survey on noise in multi-agent reinforcement learning with emergent communication.
- Developed autoencoder-based communication reliability scoring, and a graph-attention remediation mechanism weighted by a fusion of message reliability and usefulness.

Security Analyst Blockchain.com, fully remote **10/2021 – 12/2023**

Part-time alongside doctoral studies, within the security and fraud analysis team of a major cryptocurrency exchange.

- Detected fraudulent and anomalous transaction patterns by combining on-chain and off-chain sources in Python and SQL, and investigated compromised accounts.
- Identified emerging threat patterns across multiple data sources and contributed to the design of counter-measures.

Database Marketing Analyst eBay Inc., Czechia **09/2018 – 05/2021**

- Owned end-to-end execution and analytics for international campaigns within the Global Marketing Operations team.
- Engineered database segmentations basen on customer scores derived from platform propensity models, defining and scaling target audiences for global marketing campaigns.
- Coordinated with international stakeholders to ensure campaign performance metrics matched core business objectives.

Principal Data Analyst and Data Scientist Maternia s.r.o., Czechia **07/2016 – 08/2018**

Sole data specialist for an online optical retailer trading in 16 countries, with end-to-end ownership from data acquisition to models, dashboards and recommendations.

- Evaluated campaign uplift by forecasting a counterfactual sales path from pre-campaign data and comparing it with actual revenue.
- Replaced spreadsheet-based supplier ordering with LSTM forecasts of monthly purchase volume, reducing unsold stock.
- Guided webstore redesign by applying principal components analysis to an e-commerce satisfaction

survey, mapping the primary design features driving user evaluation.

- Recovered conversion-funnel drop-off across the 16 webstores by joining clickstream to transactional data; designed interventions (targeted discounts vs. recommendation) based on exit stage.
- Built a competitor price-intelligence tool that scraped competitor prices and proposed a price from the company pricing model.
- Increased customer retention by building logistic regression churn scores to drive targeted retention offers, supported merchandising with customer-feedback sentiment analysis and applying ANOVA to map product attributes to target customer segments.

Earlier experience

- **Project Coordinator, SE Europe, Thomson Reuters (02/2011 - 08/2014).** Coordinated the regional specialist team for a secure messaging and collaboration platform, managing deployment, performance analytics and user enablement.

SELECTED PORTFOLIO PROJECTS

Code, executed notebooks: github.com/anastasioskontogiorgis · Personal Portfolio with written analyses: <https://anastasioskontogiorgis.github.io>

- **Campaign uplift & causal impact:** Built a structural time-series model using uncontaminated control products to isolate true retail campaign lift; successfully recovered a +4.62% effect against a sealed +4.21% injected baseline, proving a naive before-and-after comparison (+44.8%) inflated results.
- **Semantic book recommender:** Developed a live content-based recommender over a 15,000-book catalogue using sentence-transformer embeddings; ran a controlled ablation study that successfully isolated data vs. model contributions, boosting top-5 subject relevance from 0.128 to 0.292.
- **Production Phishing Detection Service:** Containerised a FastAPI text classifier with calibrated probabilities and custom drift monitoring; audited the corpus to eliminate a 0.5% data leakage path and used a contemporary test set to expose an out-of-distribution accuracy drop from 0.980 to 0.833.
- **Ethereum Fraud & Data Forensics:** Executed non-parametric testing and multivariable logistic regression to audit blockchain address behavior; uncovered systemic data flaws that inflated fraud detection rates by 77% and engineered a corrected baseline to enable reliable predictive modeling.
- **NoisyGraph emergent-communication framework:** Designed a channel-robust multi-agent RL architecture for coordination under noisy communication constraints, resulting in three academic papers.
- **UAV Emergent-Communication Swarm:** Extended the NoisyGraph architecture to a simulated UAV swarm; diagnosed early policy-update failures and engineered a staged reward curriculum that recovered dispersed terrain coverage to a 73.2% peak.

EDUCATION

PhD in Artificial Intelligence	Trinity College Dublin, Ireland	2021 - 2026
• Thesis (submitted 04/2026, under examination): Channel-Robust Emergent Communication: Mitigating Complex Network Distortions via Reliability-Aware Graph Attention. • Robustness of learned communication between reinforcement-learning agents under message loss, delay, reordering and additive noise, applied to connected autonomous vehicles at unsignalised intersections.		
MSc in Data Science	University of Glasgow, United Kingdom	2015 - 2016
BSc (Hons) in Computer Science	University of Portsmouth, United Kingdom	2007 - 2010
Postgraduate Course in Applied Data Science	TU Dublin, Ireland	2024 - 2025
• Taken alongside the PhD, advanced coursework in statistics, deep learning, text mining, and data pipelines.		

PUBLICATIONS

- **Conference:** "CAV Unsignalised Intersection Crossing with Unreliable Emergent Communication." *ATT'24: Workshop on Agents in Traffic and Transportation*, Oct 2024.
- **Journal:** "CAV Unsignalised Intersection Crossing with Unreliable Emergent Communication: A Comprehensive Analysis under Diverse Disruptions". *European Journal on Artificial Intelligence, Special Issue*, final revision under review.
- **Journal:** "Noise in Multi-Agent Reinforcement Learning and Emergent Communication: A Systematic Taxonomy and Survey." *Autonomous Agents and Multi-Agent Systems*, under review.

LANGUAGES

Greek (native) · English (full professional proficiency) · Czech (beginner)